Sensor Fusion Task 2: Image Fusion

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1 Goal

Image fusion is the method of combining data from multiple to variety of one image that helps increases information content, visual clarity, tones down noise and adds sensor knowledge together. Faster Feature Extraction: making it faster by optimizing for feature extraction, which is useful for discipline-specific fields like remote sensing, medical imaging and defense which enables advance analysis and decision-making across all conceivable use-cases.

2 Methodology

This project explores the classification of shapes (pentagon, circle, square, triangle) using a convolutional neural network (CNN), specifically the LeNet-5 architecture. To improve classification accuracy, we implement two fusion strategies: low-level fusion and high-level fusion. These strategies integrate different types of images to leverage information regarding its shape.

LeNet-5 is a well-established CNN architecture known for its simplicity and effectiveness in image classification tasks. It consists of the following layers:

- preprocess_images(images):This function takes a set of images and resizes each image to 32x32 pixels to ensure uniform input dimensions for the CNN.

- ImageDataGenerator(rescale=1.0/255.0):This function rescales pixel values to a range of [0, 1], which helps in faster convergence during training.

- build_lenet_model(input_shape, num_classes):This function creates a LeNet CNN model, which consists of convolutional layers, pooling layers, and fully connected layers.

- train_model(model, train_images, train_labels, val_images, val_labels, epochs):This function trains the CNN model for a specified number of epochs using the training and validation datasets.

Fusion Strategies

- **low_level_fusion(datasets)**:This function concatenates multiple image datasets along the channel dimension to create a fused dataset.

  def low_level_fusion(datasets): return np.concatenate(datasets, axis=-1)

- **high_level_feature_extraction(model, images)**:This function creates a feature extractor model from the given CNN and extracts feature maps from the input images.

def high_level_feature_extraction(model, images):
feature_extractor = Model(inputs=model.input,
outputs=model.get_layer('flatten').output)
return feature_extractor.predict(images)

- Combine_high_level_features(features_list): This function concatenates feature maps extracted from different CNN models along the feature dimension.

- Evaluate_model(model, images, labels): This function computes the accuracy of the CNN model on the test dataset.

def evaluate_model(model, images, labels): return model.evaluate(images, labels)

- Generate_confusion_matrix(model, images, labels): This function calculates and plots the confusion matrix to analyze the classification performance.

- Save weights: torch.save(model.state_dict(), 'final_model_without_fusion.pth') , torch.save(model.state_dict(), 'final_model_low_level_fusion.pth') , torch.save(model.state_dict(), 'final_model_high_level_fusion.pth')

3 Evaluation

To assess the effectiveness of the low-level and high-level fusion strategies, I conducted evaluations using the ShapeDatasetLowLevelFusion dataset. The evaluation methodology is summarized as follows:

- **Model Training**: (1)Training Protocol– Both fusion strategies involved training the LeNet model for 20 epochs with a learning rate of 0.01. (2)Performance Metrics– Classification accuracy and confusion matrices were used to evaluate model performance.

- **Results**: (1)Low-Level Fusion– The confusion matrix indicated difficulties in distinguishing certain shapes, likely due to pixel-level averaging which can blur distinctive features.(2)High-Level Fusion–The confusion matrix showed improved classification performance across all shape categories, indicating that high-level feature extraction and combination provide more meaningful representations.

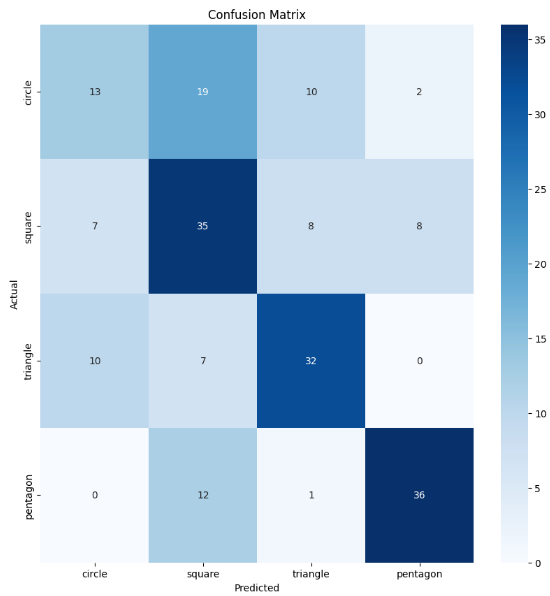

Figure 1: LeNet without Fusion (img 1)

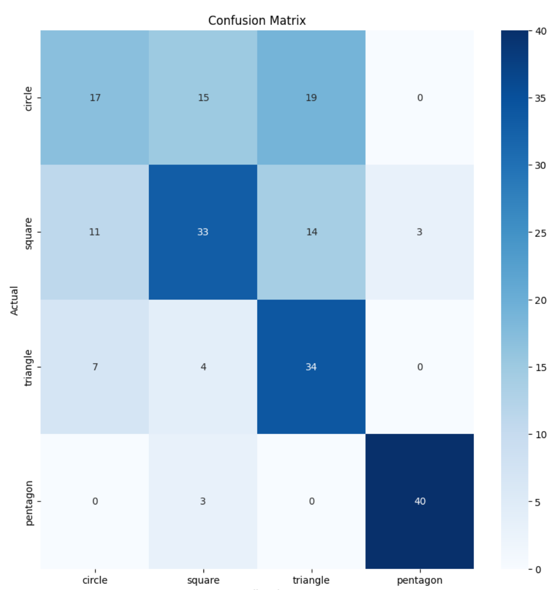

Figure 2: LeNet without Fusion (img 2)

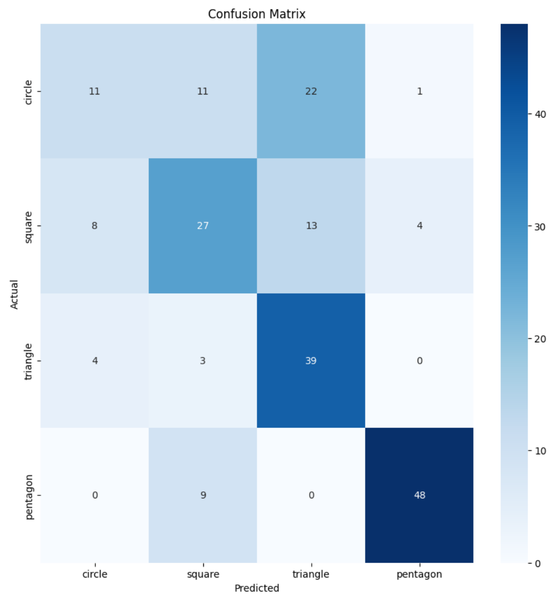

Figure 3: LeNet without Fusion (img 3)

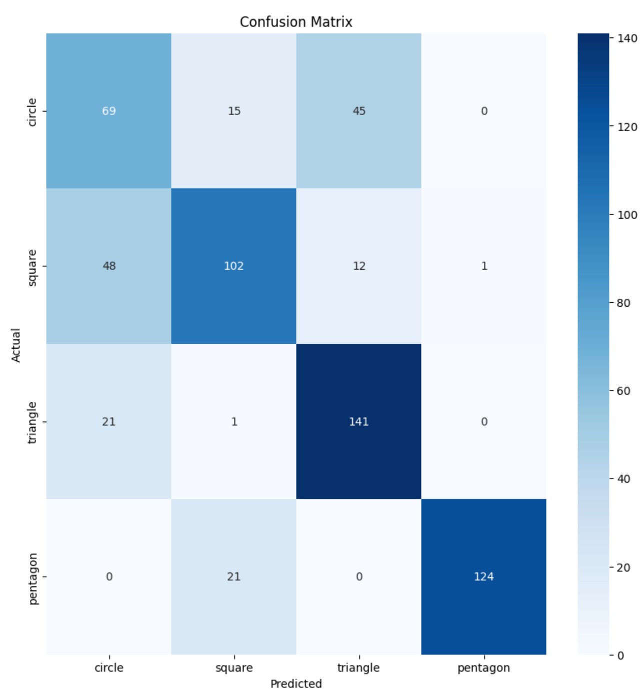

Figure 4: Low level fusion

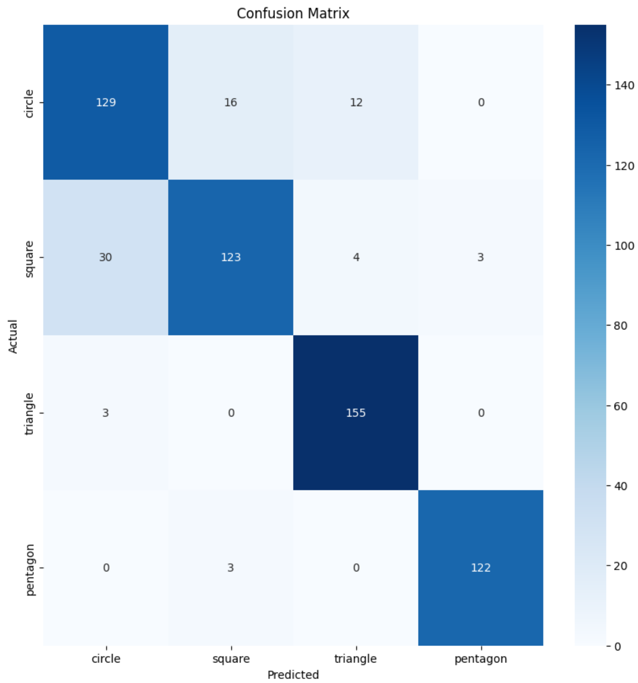

Figure 5: High level fusion

Configuration	**Accuracy (%)**
Without Fusion - img 1	58.0
Without Fusion - img 2	62.0
Without Fusion - img 3	62.5
Low-Level Fusion	72.67
High-Level Fusion	88.17

4 Conclusion

- Comparative Analysis:High-level fusion showed a more considerable improvement over low-level fusion, suggesting that combining abstract features extracted from multiple images is more effective than pixel-level concatenation. This could be due to the ability of deep features to capture more meaningful patterns.
- Low-Level Fusion Strategy: Combines raw pixel values from multiple images and creates a single fused image by averaging pixel values.
- High-Level Fusion Strategy:Extracts high-level features using a pre-trained CNN with combining feature maps from multiple images and used a feature extractor to obtain and concatenate feature representations.